# UPDATED MECHANICAL AND LINKAGE DESIGN COMPUTATIONS FOR AUTOMATED PIGPEN SCRAPER (MODEL 3)

## 1. Canonical Design Input Parameters and Linkage Specifications

This updated report incorporates the crank-slider linkage and motor-gearbox drive mechanism that actuates the linear reciprocating scraper assembly. The parameters combine CAD geometric dimensions with mechanical force and kinematic equations.

| Parameter Name | Design Symbol / Value | Engineering Units |
|---|---|---|
| Total Scraper Assembly Span | W_span = 1984.00 mm = 1.984 m | Meters (m) |
| Central Carriage Base Width | W_carriage = 1094.00 mm = 1.094 m | Meters (m) |
| Angled Wing Blade Extension | L_wing = 420.00 mm = 0.420 m | Meters (m) |
| Longitudinal Travel Distance (Stroke) | D = 2330.20 mm = 2.3302 m | Meters (m) |
| Scraper Blade Frame Height | H_blade = 129.71 mm = 0.12971 m | Meters (m) |
| Ratchet Slot Length | l_slot = 40.00 mm = 0.040 m | Meters (m) |
| Ratchet Tooth / Wall Thickness | t_tooth = 10.00 mm = 0.010 m | Meters (m) |
| Ratchet Pitch / Tooth Indexing Step | p = 50.00 mm = 0.050 m | Meters (m) |
| Forward Travel Stroke Time | t_forward = 10.0 s | Seconds (s) |
| Total Operational Cycle Time | T_c = 32.0 s | Seconds (s) |
| Design Manure Accumulation Mass | m_pile = 25.0 kg | Kilograms (kg) |
| Scraper Assembly Structure Mass | m_scraper = 35.0 kg | Kilograms (kg) |
| Wet Swine Manure Density | rho = 1000.0 kg/m³ | kg/m³ |
| Blade-Concrete Dynamic Friction Coefficient | mu = 0.40 | Dimensionless |
| Manure Surface Adhesion Shear Stress | tau_adhesion = 150.0 Pa | Pascals (N/m²) |
| Blade Floor Contact Width | b_contact = 50.0 mm = 0.050 m | Meters (m) |
| Crank Radius (Half Pawl Stroke) | r_crank = 25.00 mm = 0.025 m | Meters (m) |
| Gearbox Mechanical Efficiency | eta_gearbox = 0.85 (85%) | Dimensionless |

| | | |
|---|---|---|
| Crank-Slider Linkage Mechanical Efficiency | eta_linkage = 0.90 (90%) | Dimensionless |
| Motor Electrical-Mechanical Efficiency | eta_motor = 0.80 (80%) | Dimensionless |
| Scraper Motor Power Rating | P_rated = 1.50 HP = 1.11855 kW | Horsepower / kW |
| Scraper Duty Cycle Fraction | D_m = 0.625 (62.5%) | Dimensionless |
| Standard Operational Cleaning Run Time | t_run = 15.0 min | Minutes (min) |

## 2. Kinematic and Crank-Slider Mechanism Computations

This section calculates the linear speed of the scraper carriage, crank radius, gearbox output rotational speed, and peak pawl velocity.

**Equation 2.1: Scraper Average Linear Forward Velocity (v_avg):**

*Formula:* $v_avg = D / t_forward$

$v_avg = 2.3302\ \text{m} / 10.0\ \text{s}$

$v_avg = 0.23302\ \text{m/s}$

**Equation 2.2: Ratchet Tooth Pitch (p):**

*Formula:* $p = l_slot + t_tooth$

$p = 0.040\ \text{m} + 0.010\ \text{m}$

$p = 0.050\ \text{m} = 50.00\ \text{mm}$

**Equation 2.3: Total Ratchet Indexing Steps per Stroke (N_steps):**

*Formula:* $N_steps = D / p$

$N_steps = 2.3302\ \text{m} / 0.050\ \text{m}$

$N_steps = 46.604\ \text{steps} \approx 47\ \text{ratchet indexing cycles}$

**Equation 2.4: Crank Radius for Single-Pitch Pawl Actuation (r_crank):**

*Formula:* $r_crank = p / 2$

r_crank = 0.050 m / 2

r_crank = 0.025 m = 25.00 mm

**Equation 2.5: Required Gearbox Rotational Speed (N_gearbox):**

*Formula: N_gearbox = (N_steps / t_forward) × 60*

N_gearbox = (46.604 rev / 10.0 s) × 60 s/min

N_gearbox = 279.624 RPM

**Equation 2.6: Gearbox Angular Velocity (omega_gearbox):**

*Formula: omega_gearbox = (2 × pi × N_gearbox) / 60*

omega_gearbox = (2 × 3.14159 × 279.624 RPM) / 60

omega_gearbox = 29.282 rad/s

**Equation 2.7: Peak Linear Speed of the Reciprocating Pawl (v_peak):**

*Formula: v_peak = omega_gearbox × r_crank*

v_peak = 29.282 rad/s × 0.025 m

v_peak = 0.73205 m/s

## 3. Volumetric Waste Capacity and Spillover Computations

This section evaluates the maximum holding capacity of the scraper blade before manure overflows over the frame height.

**Equation 3.1: Maximum Triangular Waste Cross-Sectional Area (A_max):**

*Formula: A_max = 0.5 × (H_blade)^2*

A_max = 0.5 × (0.12971 m)^2

A_max = 0.5 × 0.016825 $m^2$

A_max = 0.0084123 $m^2$

**Equation 3.2: Maximum Volumetric Waste Capacity (V_max):**

*Formula: $V_{max} = A_{max} \times W_{span}$*

$V_{max} = 0.0084123\ m^2 \times 1.984\ m$

$V_{max} = 0.016690\ m^3 = 16.690\ Liters$

**Equation 3.3: Maximum Waste Mass Capacity Before Spillover (m_max):**

*Formula: $m_{max} = V_{max} \times rho$*

$m_{max} = 0.016690\ m^3 \times 1000.0\ kg/m^3$

$m_{max} = 16.690\ kg$

**Equation 3.4: Total Swept Floor Area per Stroke (A_pen):**

*Formula: $A_{pen} = W_{span} \times D$*

$A_{pen} = 1.984\ m \times 2.3302\ m$

$A_{pen} = 4.62312\ m^2$

**Equation 3.5: Maximum Single-Pass Uniform Floor Layer Thickness (h_layer,max):**

*Formula: $h_{layer,max} = V_{max} / A_{pen}$*

$h_{layer,max} = 0.016690\ m^3 / 4.62312\ m^2$

$h_{layer,max} = 0.003610\ m = 3.610\ mm$

## 4. Dynamic Resistance Force and Crank Torque Computations

This section evaluates the dynamic resistance forces opposing carriage motion and calculates the resulting peak torque on the gearbox output shaft.

**Equation 4.1: Manure Accumulation Pushing Force (F_pile):**

*Formula: $F_{pile} = m_{pile} \times g$*

$F_{pile} = 25.0\ kg \times 9.81\ m/s^2$

$F_{pile} = 245.250\ N$

**Equation 4.2: Total Vertical Normal Force on Pen Floor (N_total):**

*Formula:* $N_{total} = (m_{scraper} + m_{pile}) \times g$

$N_{total} = (35.0\ kg + 25.0\ kg) \times 9.81\ m/s^2$

$N_{total} = 60.0\ kg \times 9.81\ m/s^2$

$N_{total} = 588.600\ N$

**Equation 4.3: Mechanical Floor Friction Force (F_friction):**

*Formula:* $F_{friction} = mu \times N_{total}$

$F_{friction} = 0.40 \times 588.600\ N$

$F_{friction} = 235.440\ N$

**Equation 4.4: Blade Base Floor Contact Footprint Area (A_contact):**

*Formula:* $A_{contact} = W_{span} \times b_{contact}$

$A_{contact} = 1.984\ m \times 0.050\ m$

$A_{contact} = 0.09920\ m^2$

**Equation 4.5: Surface Adhesion Shear Force (F_adhesion):**

*Formula:* $F_{adhesion} = tau_{adhesion} \times A_{contact}$

$F_{adhesion} = 150.0\ N/m^2 \times 0.09920\ m^2$

$F_{adhesion} = 14.880\ N$

**Equation 4.6: Total Dynamic Resistance Force (F_total):**

*Formula:* $F_{total} = F_{pile} + F_{friction} + F_{adhesion}$

$F_{total} = 245.250\ N + 235.440\ N + 14.880\ N$

$F_{total} = 495.570\ N$

**Equation 4.7: Peak Gearbox Output Shaft Torque (tau_peak):**

*Formula: tau_peak = F_total × r_crank*

tau_peak = 495.570 N × 0.025 m

tau_peak = 12.389 N·m

## 5. Power Sizing and Motor Safety Factor Verification

This section determines the average and peak mechanical power, required input electrical power, and verifies the motor safety factor under peak linkage torque.

**Equation 5.1: Average Mechanical Power Output (P_avg,mech):**

*Formula: P_avg,mech = F_total × v_avg*

P_avg,mech = 495.570 N × 0.23302 m/s

P_avg,mech = 115.478 W = 0.11548 kW

**Equation 5.2: Peak Mechanical Power Output (P_peak,mech):**

*Formula: P_peak,mech = F_total × v_peak*

P_peak,mech = 495.570 N × 0.73205 m/s

P_peak,mech = 362.782 W = 0.36278 kW

**Equation 5.3: Mechanical Drive Transmission Efficiency (eta_drive):**

*Formula: eta_drive = eta_gearbox × eta_linkage*

eta_drive = 0.85 × 0.90

eta_drive = 0.765 (76.5%)

**Equation 5.4: Total Drive and Motor System Efficiency (eta_system):**

*Formula: eta_system = eta_drive × eta_motor*

eta_system = 0.765 × 0.80

eta_system = 0.612 (61.2%)

**Equation 5.5: Required Electrical Input Power at Average Load (P_req,avg):**

*Formula: P_req,avg = P_avg,mech / eta_system*

P_req,avg = 115.478 W / 0.612

P_req,avg = 188.689 W = 0.18869 kW

**Equation 5.6: Required Electrical Input Power at Peak Sinusoidal Load (P_req,peak):**

*Formula: P_req,peak = P_peak,mech / eta_system*

P_req,peak = 362.782 W / 0.612

P_req,peak = 592.781 W = 0.59278 kW

**Equation 5.7: Required Horsepower at Peak Load (HP_req,peak):**

*Formula: HP_req,peak = P_req,peak / 745.7*

HP_req,peak = 592.781 W / 745.7 W/HP

HP_req,peak = 0.79493 HP

**Equation 5.8: Motor Capacity Safety Factor at Peak Load (SF_peak):**

*Formula: SF_peak = P_rated_HP / HP_req,peak*

SF_peak = 1.500 HP / 0.79493 HP

SF_peak = 1.887

## 6. Operational Energy Consumption Computations

This section calculates the total electrical energy consumed per operational cleaning cycle based on duty cycle and motor rating.

**Equation 6.1: Active Scraper Operation Duration (t_active):**

*Formula: $t_active = t_run \times D_m$*

t_active = 15.0 min × 0.625

t_active = 9.375 min = 0.15625 hours

**Equation 6.2: Total Electrical Energy Consumed per Run (E_scraper):**

*Formula: E_scraper = P_rated_kW × D_m × (t_run / 60)*

E_scraper = 1.11855 kW × 0.625 × (15.0 min / 60.0 min/h)

E_scraper = 1.11855 kW × 0.625 × 0.250 h

E_scraper = 0.17477 kWh ≈ 0.175 kWh